

sol3_research_deep_agents. White paper

A question in, a cited brief out. A topic in, an evidence-backed white paper out.

LangChain Deep Agents. Two entry points. One role table.

This folder has no `HOW_TO_RUN.md`. Read `SPEC.md`, `DESIGN_DOC.md`, and `Taskfile.yml`.

Saturday Lab 3 still fills two functions. This folder writes `whitepaper.md`.

Two entry points. Same folder.

```
task brief -- --question "sqlalchemy nullable datetime column" --backend fixture
task paper      # nine stages, fixture research
task live       # backend auto. Needs task setup.
```

The brief is the small version of the paper. Both plan questions, search through one tool boundary, and refuse to ship anything uncited.

Setup

```
cd solutions/sol3_research_deep_agents
echo 'ANTHROPIC_API_KEY=sk-ant-...' >> ../../.env
task setup
```

`.venv` plus Deep Agents. Also clones `imagen-diagrams v0.2.0` and `image-gen v2.1.0` into `.cache/`.

`task test`, `task table`, `task checks`, and `task brief` need none of that.

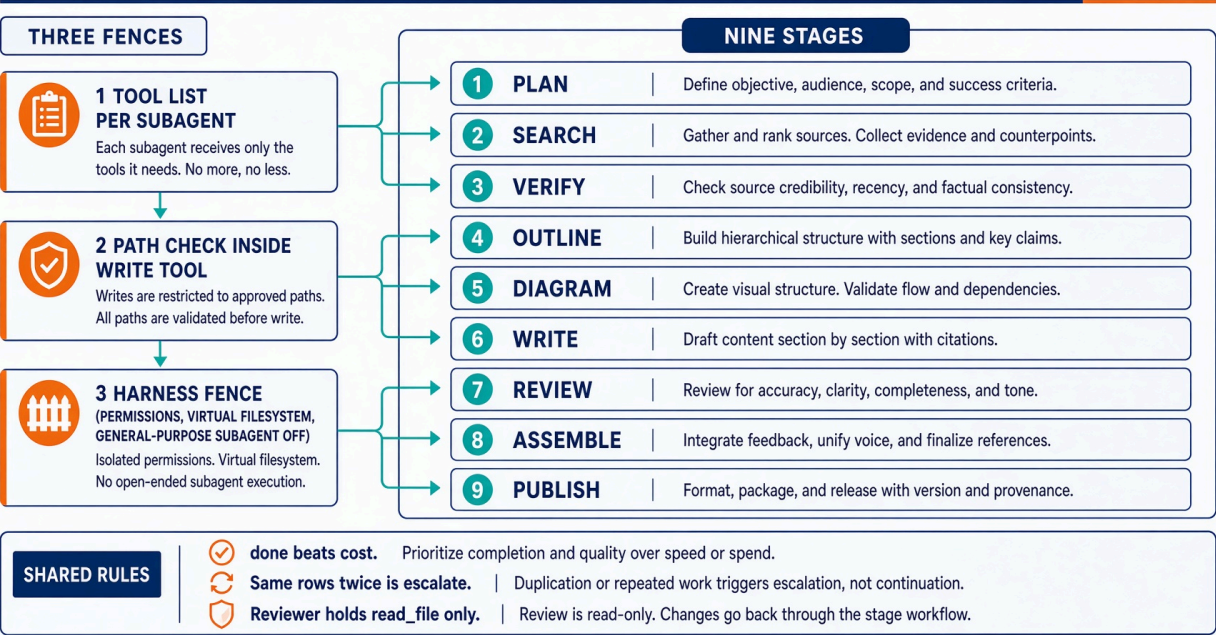
Scripts with no model

```
task table          # paper cast. Reviewer writes must print no
task checks         # evidence, paper_check, diagrams, publish, mcp_tools
task test
```

If the reviewer prints `yes`, stop.

Starting architecture

Deep Agents white paper. Nine stages. Three fences.



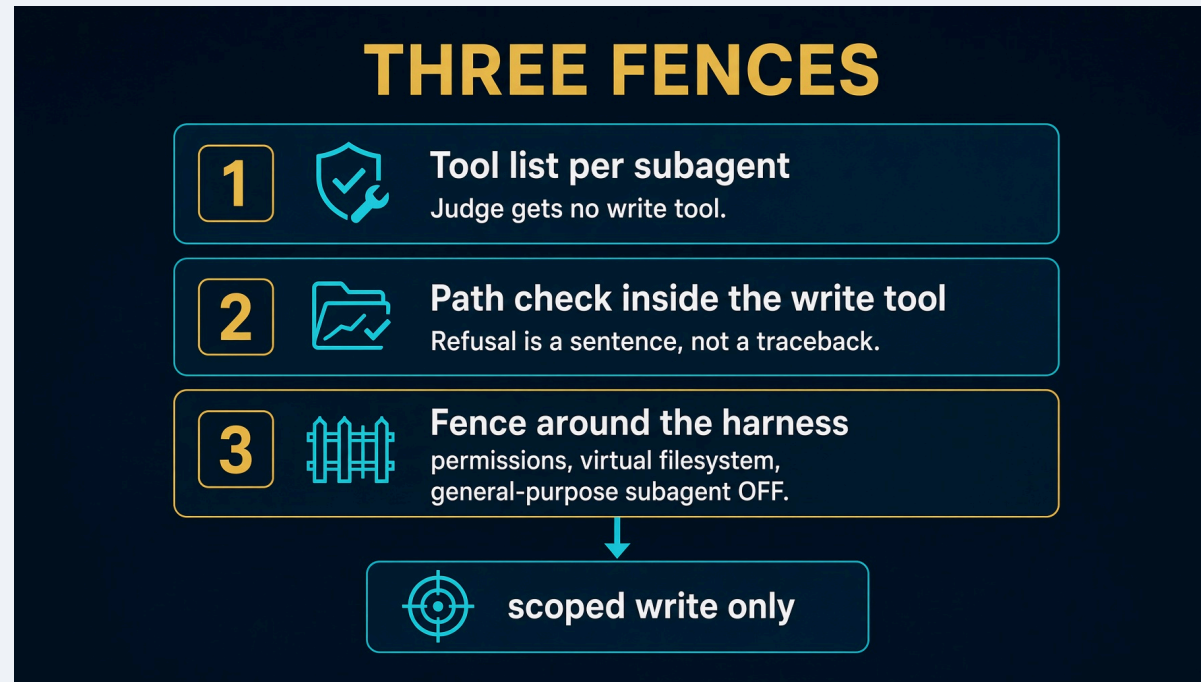
See also [docs/diagrams/architecture.svg](#).

The paper cast

role	writes	scope
orchestrator	no	nothing
planner	yes	plan.json
researcher	no	nothing
verifier	yes	evidence/** denied: paper/**
diagrammer	yes	diagrams/*.mmd, *.puml
writer	yes	brief.md, paper/**, work/research/**
reviewer	no	nothing

Arole that would hold the same tools as its neighbour is not a role. It is a prompt, and it belongs in a skill.

Three fences. People skip the third.



Do not shadow `read_file`. That is how `/skills/` goes silent.

Nine stages. Two of them are Python.

#	STAGE	MODEL?	GATE
1	plan	yes	3 to 8 questions, each with a check
2	search	yes	every claim names a source that resolves
3	verify	yes	every important claim has a decided truth state
4	outline	yes	every body section names claim ids that exist
5	diagram	yes	at most 12 nodes, every figure has alt text
6	write	yes	a section cites only the numbers its claims support
7	review	yes	every rubric row passes
8	assemble	no	every hard gate in <code>paper_check</code>
9	publish	no	the gates passed. Opt-in only

Three exits. Done first.

Checked before every stage:

EXIT	WHEN
done	stage 8 finished and every hard gate is green
cost	the money budget is spent
max turns	a stage exhausted its retries

Done beats a spent budget. A run that finished and then noticed it was over its cap did finish.

Cost is checked before every model call, not between stages. A spent budget is never a retry.

Saturday brief is still here

```
def plan_questions(question: str) -> list[str]:  
    return [question, f"{question} common mistake", f"{question} how to verify"]  
  
def check_brief(body, sources):  
    return brief.check(body, sources)
```

Four arithmetic rows: `has_sources`, `grounded`, `cited`, `style`.

```
task brief -- --question "sqlalchemy nullable datetime column"
```

That is the Lab 3 answer. The paper is the take-home grown from it.

Figures fail closed

Mermaid and PlantUML are source. They are not publication formats.

`imagen-diagrams v0.2.0` renders `*_imagen.png` and judges it. `image-gen v2.1.0` is cover art only.

A missing backend writes `<stem>_imagen.prompt.txt` and exits 2. The paper never substitutes SVG.

Publish. Four load-bearing rules.

1. **One gist per paper, forever.** Id in `gist-ids.tsv`.
2. **Secret, never public.** Unlisted. The URL is the credential.
3. **Figures inline.** A gist is flat.
4. **The gate comes first.** `push` refuses when `gates.json` is red.

```
task paper          # never publishes
task publish -- --slug ... --dry-run --out /tmp/staged
```

Testing skill

`.agents/skills/e2e-test-research-report/`

Default lane is the Agent SDK twin. Point it at this folder only when you name it and the tasks exist.

Read `SPEC.md` and `Taskfile.yml` first. `task test` and `task checks` before any live spend.

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Troubleshooting

SYMPTOM	FIX
Reviewer writes <code>yes</code>	<code>read_file</code> only
Agent writes anywhere	turn the general-purpose subagent off
Traceback retry storm	return a refusal sentence
New gist every run	one id per slug in <code>gist-ids.tsv</code>
Published a red paper	<code>push</code> must refuse
Saturday brief missing	<code>--question</code> , not <code>--paper</code>
No <code>HOW_TO_RUN.md</code>	use <code>SPEC.md</code> + <code>Taskfile.yml</code>

Recap

A brief for Saturday. A paper for take-home. Same exits.

1. Two entry points. One table.
2. Three fences. The third is the one people skip.
3. Assemble and publish are Python.
4. Done beats cost.
5. Claims outlive the paper.

A role that holds the same tools as its neighbour is a prompt.